Windows Worker Node Labs

Joining, scheduling and troubleshooting Windows nodes

Hybrid Linux + Windows Kubernetes · Enterprise

Format: PDF · ~12 pages
Audience: Beginner → Engineer → Enterprise
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Concepts you must know first

Windows nodes run **kubelet.exe** and **kube-proxy.exe** as Windows Services. Networking is handled by HNS (Host Networking Service), not iptables.

Containers use **process isolation** (lightweight, kernel-shared) or **Hyper-V isolation** (a tiny VM per container — required for multi-tenant or stricter security).

Windows images must match the host build number — a Windows Server 2022 node cannot run a Windows Server 2019 container in process isolation.

Lab — Join a Windows node to a Linux cluster

1. Install containerd on the Windows node:

```
curl.exe -LO https://github.com/containerd/containerd/releases/download/v1.7.20/containerd-1.
7.20-windows-amd64.tar.gz
tar xvf containerd-1.7.20-windows-amd64.tar.gz -C C:\Program Files\containerd
containerd.exe --register-service
Start-Service containerd
```

2. Install kubelet and join with the bootstrap token:

```
Invoke-WebRequest -Uri https://dl.k8s.io/v1.30.0/bin/windows/amd64/kubelet.exe -OutFile
C:\k\kubelet.exe
Invoke-WebRequest -Uri https://dl.k8s.io/v1.30.0/bin/windows/amd64/kubeadm.exe -OutFile
C:\k\kubeadm.exe
kubeadm.exe join 10.0.0.10:6443 --token <token> --discovery-token-ca-cert-hash sha256:<hash>
```

3. Verify from the Linux control plane:

```
kubectl get nodes -L kubernetes.io/os
# Look for STATUS=Ready, OS=windows
```

Lab — Schedule a Windows IIS pod

```
apiVersion: apps/v1
kind: Deployment
metadata: {name: iis}
spec:
replicas: 2
selector: {matchLabels: {app: iis}}
template:
metadata: {labels: {app: iis}}
spec:
nodeSelector: {kubernetes.io/os: windows}
tolerations: [{key: os, value: windows, effect: NoSchedule}]
containers:
- name: iis
image: mcr.microsoft.com/windows/servercore/iis
ports: [{containerPort: 80}]
```

Troubleshooting Windows nodes

Symptom	Diagnose with	Common fix

Node NotReady	Get-Service kubelet, containerd	Restart-Service kubelet
Image won't pull	ctr -n k8s.io images pull <img>	Match Windows build number
Pod has no IP	Get-HnsNetwork; ipconfig /all	Restart HNS: Restart-Service hns
Service IP unreachable	Get-Service kube-proxy; vfprules	Restart-Service kube-proxy